Gains on Synthetic Occlusion Are Not Enough: Transfer of Occlusion Augmentation to Real Occlusion Depends on the Dose in Person Detection

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***Abstract*—Synthetic occlusion augmentation is used to make person detectors robust to partial visibility, yet its benefit is usually measured on the same artificial occlusion. We ask whether a gain on synthetic occlusion is enough to expect a gain on real occlusion annotated in other datasets. In 21 controlled blocks (133 runs), a baseline and an augmented model share data and hyperparameters; the blocks vary the occluder, the detector (six, one-stage, two-stage and transformer), the domain and the augmentation dose, and are measured on a synthetic control and on three real-occlusion sources, by day and by night, with hypotheses registered internally with a cryptographic fingerprint before training. All 20 blocks with a control improve on it (+2.6 to +60.9 AP@0.5 points). With the usual dose and equal exposure (12 blocks), a mixed model estimates at most +1.1 points per source and rules out a 2-point gain on all three. Only a high dose with Faster R-CNN improves two real sources, and it does so equally for unoccluded people. Passing a synthetic control is not enough to claim robustness to real occlusion.**

***Index Terms*—person detection, occlusion, data augmentation, robustness, low light, hypotheses registered with a cryptographic fingerprint**

I. Introduction

Person detection with RGB cameras underpins video surveillance and mobile robotics, which also operate at night. In that setting, occlusion, a major cause of pedestrian-detection failures [**?**], and low light [**?**] coincide. Those affected are hidden people the detector misses and whoever relies on it to monitor or avoid a collision. In our data, the baselines lose 0.39 to 0.62 AP@0.5 on occluded people relative to visible ones, by day and by night (Section IV).

A widespread remedy is *synthetic occlusion augmentation*: during training, parts of the person are erased or covered [**?**], [**?**], [**?**]. Its benefit is usually measured on the same artificial occlusion or, when real occlusion is used, within the training dataset (CityPersons, CrowdHuman [**?**]); rarely on genuinely occluded people in other datasets. If the synthetic gain does not carry over, synthetic benchmarks overstate robustness and a surveillance system would be deployed with a robustness it lacks. In Peru, municipal security services had 14,722 operational video-surveillance cameras in 2023 [**?**]; with no public dataset of people occluded at night recorded in the country that we know of, those systems would rely on models trained on other data, and this study, like them, uses only foreign datasets.

Gap. In the literature reviewed (Section II) we found no study that (i) measures the effect on annotated real occlusion with a synthetic control confirming that the augmentation was learned, (ii) with several real sources, (iii) tests alternative explanations (occluder, detector, domain, data volume, exposure, dose) with controlled experiments and (iv) fixes hypotheses and rules before seeing the results.

Question, objectives and contribution. The general objective is to determine whether synthetic occlusion augmentation that improves a synthetic control also improves the detection of genuinely occluded people. The specific objectives are to measure the baseline's degradation under occlusion (OE1, AP@0.5 by stratum); the effect of augmentation (V3−V1) by occluder and dose (OE2), detector (OE3), and training domain and data (OE4); and to explore its mechanism (OE5: recall, specificity and occluder realism)[1]. We contribute (1) controlled evidence in 21 blocks with five kinds of occluder, six detectors and three doses; (2) a global mixed-model estimate and a specificity analysis (occluded versus visible) showing that passing the synthetic control is not enough, and that the only transfer that holds, at a high dose, is not specific to occlusion; and (3) a minimum protocol (positive control, V1 and V3 on the same machine, fixed epochs, 5 seeds, several real sources, specificity and a rule fixed in advance), met in full only by H18a; each block costs 0.5–2.6 GPU hours on an RTX 4090.

II. Related work and explanatory framework

Detection under occlusion and low light. Pedestrian detectors tackle occlusion through architecture and evaluate within one benchmark: by parts or visible boxes [**?**], [**?**], [**?**] or by fusing levels with Random Erasing [**?**]; at night, a YOLOX with domain transfer raises mAP by 5.9 points without stratifying by occlusion [**?**].

Augmentation as a training signal and as a test. Random Erasing [**?**] inspired pedestrian variants evaluated within CityPersons or CrowdHuman [**?**], [**?**]; Copy-Paste [**?**] and

[1]The mapping between objectives, method, evidence and pending work, the literature search and all material cited as "supplement" are in the supplementary material.

A-Fast-RCNN [**?**] paste objects or occlude features without measuring carry-over to other datasets. Synthetic-corruption benchmarks [**?**] measure what a detector withstands, not whether gains carry over; Taori et al. find that robustness to synthetic shifts barely carries over to natural shifts [**?**], Hendrycks et al. that it sometimes does and sometimes does not [**?**], and Mintun et al. that the gain depends on the similarity between augmentation and test [**?**]. Kassaw et al. find that synthetic occlusion *ranks* already-trained classifiers much as real occlusion does [**?**], and Pytel et al. that occlusion augmentation does not solve occlusion in pose estimation [**?**].

Alternatives and explanatory framework. Compositional networks represent objects by parts [**?**], [**?**]; WALT learns amodal representations from real occlusion [**?**]. As a framework we use shortcut learning [**?**]: a network can reduce its loss by learning the artificial occluder without learning to detect incomplete people, and pedestrian detectors already generalize poorly across datasets [**?**], [**?**]. The search started from Scopus, which yielded the five seed articles [**?**], [**?**], [**?**], [**?**], [**?**].

III. Method

Design, hypotheses and rule. The unit of analysis is the cell: one block (a V1–V3 pair) evaluated on one source. In each block, V1 and V3 share data, hyperparameters, base augmentation and machine, except H16, whose V1 are those of H15, trained on another machine of the same model; V3 adds occlusion augmentation only in training. H3 and H4 predicted that the synthetic gain would carry over to real occlusion; the later transfer hypotheses predict that it does not appear when the occluder (H9, H9b, H10, H14; OE2), the detector (H8, H12, H15, H17; OE3), the domain and data (H6, H7; OE4) or the dose (H18; OE2) change. The prediction holds (transfer *not detected*) if fewer than two of the three real sources *pass*, i.e., improve AP@0.5 by at least +2 percentage points (pp, mean over seeds) with a 95 % image-level CI that excludes zero, and the synthetic control does pass; if the control fails, the block is inconclusive. H11 predicted the opposite (a gain on the most occluded CityPersons people), H16a and H16b that the Faster R-CNN transfer would extend to other occluders, and H13 a realism ordering of the occluders (OE5). Each hypothesis, its rule and its code were registered internally with a SHA-256 cryptographic fingerprint before training its block and none was reformulated; the fingerprint proves that nothing changed after the recorded date, but there was no external registry. The programme is adaptive (Fig. 1): the question was reoriented after the first experiment, H9b–H17 were designed after seeing earlier results and H18 answers an external review. Anything marked as post hoc decides no hypothesis.

Data. ExDark [**?**], with the split of Cui et al. [**?**], has 2,658 images with people; 1,183 are used for training. Real-occlusion sources: (1) NightOwls [**?**], at night and from a vehicle, with its authors' binary occlusion attribute (530 occluded pedestrians in 2,000 validation frames); (2) CityPersons [**?**], by day, *heavy* (visibility 20–65 %, height $\geq$50 px); (3) a proxy of occlusion between people in ExDark from box overlap. This proxy is post hoc: it was adopted on 13/09/2026, after H3 and H4, because the planned manual visibility annotation was unreliable (κ 0.06–0.60 between passes), and its validity is low (Spearman 0.24 and κ 0.24 against annotated CityPersons visibility); from H6 on it was fixed before training. We therefore repeat the analysis without it (Section IV). CityPersons and NightOwls are scored with the full-body box, as in their protocols, and ExDark with the box of its annotations; a synthetically occluded person keeps its full box and pasted person silhouettes are labelled. All three datasets are used under non-commercial research licences.

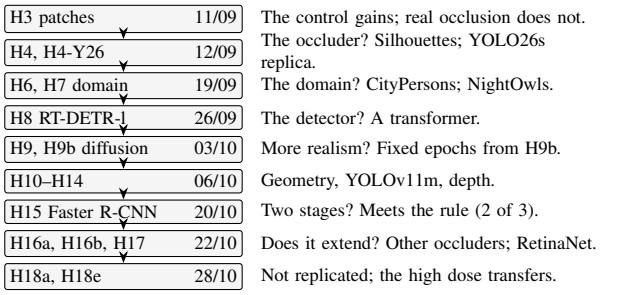


Fig. 1. Adaptive programme: each block answers an alternative explanation of the previous one (2026 dates on which the hypothesis was fixed).

Augmentation, dose and control. Occluder families: patches; silhouettes of real objects and people; diffusion (Stable Diffusion 1.5 [**?**]); silhouettes with the real visibility geometry of CityPersons; and depth (Depth Anything V2 [**?**]). The usual dose occludes an image with probability 0.5, and in it 1–3 people with coverage from 0.30 to 0.85 (for diffusion and depth, the occluded image replaces the original with probability 0.5). H18e changes only the silhouette dose: low (probability 0.25, one person, coverage 0.15–0.40) and high (1.0; 1–6 people; 0.50–0.95). The *control* applies the same generator to every person in the test set of the training domain with coverage from 0.70 to 1.00 and unseen occluders; the cross-family matrix uses the other families' controls (parameters in the supplement).

Detectors and exposure. YOLOv11s, YOLOv11m [**?**], YOLO26s [**?**] (SGD, lr 0.002, batch 16), RT-DETR-l [**?**] (AdamW, 10^{-4}) and, from torchvision with the ResNet-50 FPN backbone, Faster R-CNN [**?**] and RetinaNet [**?**] (SGD, batch 8, lr 0.01 and 0.005), pre-trained on COCO [**?**]; input and evaluation at 640 px on the long side. In H4, H4-Y26, H6 and H8 each variant was selected by validation, with *unequal* exposure (V1 at epochs 1–51, V3 at 31–89); from H9b on, V1 and V3 are trained for 30 fixed epochs (60 in H11) and evaluated with the last weights: these 14 blocks are the primary evidence. Each variant is trained with 3 seeds; H9b, H10-s and H18a with 5. In total there were 133 runs and 109 GPU hours, 79 of them in the first seven blocks (log in the supplement). Machines: a laptop with an NVIDIA GeForce RTX 5050 Laptop GPU (8 GB) and, for H11, H12 and H14–H18, a rented RTX 4090.

Evaluation. AP@0.5 of *person* by occlusion stratum, with the other strata as *ignore* regions [**?**]; it is the only metric

TABLE I

BLOCKS, HYPOTHESES AND RESULTS: CHANGE IN AP@0.5 (PP) OF V3 OVER V1, MEAN OF 3 SEEDS (5 IN H18A). EXP.: TRAINING EXPOSURE, EQUAL (E, FIXED EPOCHS, PRIMARY EVIDENCE), UNEQUAL (U) OR EARLY IN BOTH (EA). NO: NIGHTOWLS OCCLUDED ($n = 530$); CP: CITYPERSONS *heavy* ($n = 733$); EX: EXDARK PROXY, SEVERE ($n = 101$); CONTROL, 684–2,131 PEOPLE. IN BRACKETS, 95 % CI OVER IMAGES (1,000 REPLICATES). * PASSES: +2 PP WITH CI > 0. ° DOES NOT PASS WITH THE HIERARCHICAL CI (SEEDS AND IMAGES), THE UNIFORM SEED CRITERION FOR EVERY CELL. VERDICTS: *Not detected (k of 3)*, FEWER THAN 2 SOURCES PASS (THE PREDICTION HOLDS); *Detected*, 2 OR 3 PASS (THE PREDICTION IS REFUTED); *Not confirmed*, IT PREDICTED A GAIN AND THERE WAS NONE; *Refuted*, IT PREDICTED TRANSFER AND THERE WAS NONE; *Inconclusive*, NO CONTROL; *Same pattern*, REPLICATION OF H4 WITH NO HYPOTHESIS OF ITS OWN; H10-S IS THE SILHOUETTE ARM OF H10, WITH NO VERDICT OF ITS OWN. † NOT REPLICATED WITH 5 SEEDS (H18A).

Block	Detector	Training	Occluder	Exp.	Control	NO	CP	EX	Verdict
H3	YOLOv11s	ExDark	Patches	Ea	+60.92*	−3.17 [−4.6; −1.7]	−0.36 [−1.1; +0.3]	−3.25 [−5.8; −0.2]	Not confirmed (predicted a gain)
H4	YOLOv11s	ExDark	Silhouettes	U	+10.63*	−16.79 [−19.6; −14.0]	−3.94 [−5.7; −2.4]	−5.02 [−10.7; +0.5]	Not confirmed (predicted a gain)
H4-Y26	YOLO26s	ExDark	Silhouettes	U	+12.01*	−14.31 [−17.4; −11.1]	−4.42 [−6.0; −2.6]	−3.51 [−8.4; +1.2]	Same pattern
H6	YOLOv11s	CityPersons	Silhouettes	U	+23.74*	−1.35 [−2.9; +0.2]	+1.82 [+0.3; +3.3]	+0.94 [−2.5; +4.7]	Not detected (0 of 3)
H7	YOLOv11s	NightOwls 20,000	Silhouettes	U	—	−1.43 [−4.0; +1.0]	−3.94 [−5.6; −2.3]	−9.15 [−13.9; −5.7]	Inconclusive (no control)
H8	RT-DETR-l	ExDark	Silhouettes	U	+10.97*	−6.98 [−8.9; −5.1]	−3.47 [−4.9; −1.8]	−2.21 [−6.2; +1.6]	Not detected (0 of 3)
H9	YOLOv11s	ExDark	Diffusion	Ea	+2.60*	−1.93 [−2.9; −0.9]	+1.23 [+0.4; +2.1]	+2.82*° [+0.4; +6.0]	Not detected (1 of 3)
H9b	YOLOv11s	ExDark	Diffusion	E	+15.23*	+0.01 [−1.0; +1.2]	+1.52 [+0.8; +2.7]	−3.03 [−6.8; +1.0]	Not detected (0 of 3)
H10	YOLOv11s	ExDark	Real geometry	E	+18.39*	−2.05 [−3.3; −0.9]	+2.37* [+1.5; +3.5]	−4.88 [−9.1; −0.8]	Not detected (1 of 3)
H10-s	YOLOv11s	ExDark	Silhouettes	E	+10.99*	−1.12 [−2.1; +0.1]	+2.26* [+1.4; +3.5]	−3.50 [−7.6; +0.1]	(H10 arm)
H11	YOLOv11s	CityPersons	Silhouettes	E	+24.66*	+1.95 [+0.9; +3.2]	+1.31 [−0.3; +2.8]	−2.44 [−6.8; +2.4]	Not confirmed (predicted a gain)
H12a	YOLOv11m	ExDark	Silhouettes	E	+14.45*	−0.39 [−1.6; +0.9]	+0.25 [−0.6; +1.3]	+3.04 [−1.1; +6.6]	Not detected (0 of 3)
H12b	YOLOv11m	ExDark	Diffusion	E	+19.04*	+0.78 [−0.4; +2.2]	+1.11 [+0.1; +2.2]	−0.56 [−4.1; +3.2]	Not detected (0 of 3)
H14	YOLOv11s	ExDark	Depth	E	+7.42*	−1.50 [−2.8; −0.4]	−0.16 [−1.2; +0.8]	+5.49*° [+1.6; +9.3]	Not detected (1 of 3)
H15	Faster R-CNN	ExDark	Silhouettes	E	+14.72*	+3.28* [+1.6; +4.8]	+2.33* [+1.0; +3.7]	−1.00 [−5.6; +3.2]	Detected (2 of 3)†
H16a	Faster R-CNN	ExDark	Diffusion	E	+20.35*	+0.92 [−0.5; +2.3]	−0.76 [−1.8; +0.5]	+0.80 [−3.5; +5.0]	Refuted (predicted transfer)
H16b	Faster R-CNN	ExDark	Depth	E	+9.63*	+0.37 [−1.0; +1.7]	+0.60 [−0.7; +2.0]	−0.45 [−6.2; +4.6]	Refuted (predicted transfer)
H17	RetinaNet	ExDark	Silhouettes	E	+6.60*	−2.11 [−3.5; −0.7]	−0.28 [−1.2; +0.7]	−3.77 [−8.5; +0.8]	Not detected (0 of 3)
H18a	Faster R-CNN	ExDark	Silhouettes	E	+14.44*	+4.26* [+2.9; +5.5]	+1.74 [+0.5; +2.9]	−2.20 [−6.1; +1.6]	Not detected (1 of 3)
H18e-l	Faster R-CNN	ExDark	Silhouettes, low dose	E	+2.94*	+2.90* [+1.7; +4.2]	+1.74 [+0.4; +2.8]	−1.91 [−5.1; +2.0]	Not detected (1 of 3)
H18e-h	Faster R-CNN	ExDark	Silhouettes, high dose	E	+25.62*	+8.59* [+6.1; +10.6]	+3.78* [+2.1; +5.3]	+0.60 [−4.1; +5.1]	Detected (2 of 3)

common to the three sources and the control. CIs by *bootstrap* over images (1,000 replicates) [**?**], which decides the rule, and hierarchical over seeds and images (2,000), which we apply to every cell as the uniform between-seed criterion. The +2 pp threshold is the smallest effect of interest: it equals, for Faster R-CNN, about 17 more occluded people detected per 1,000 at 0.1 false positives per image (V1 detects 91 on CityPersons *heavy* and 196 on NightOwls; post hoc analysis), and exceeds the between-seed SD of V1 (median 1.5 pp). In the text, cell CIs are hierarchical; Table I gives the image CIs. Not detecting transfer does not prove its absence: with a true +2 pp on all three sources the rule detects it in one block out of three (with +3 pp, in 94 %). A cell *rules out* +2 pp if the upper limit of its CI is below +2 [**?**]. As a global estimate (post hoc, added after the external review) we fit a mixed model by REML to the per-seed differences: a fixed effect per source and random effects for the block and for the seed within the block. Secondary: MR^{-2} [**?**] and AP@0.5:0.95 (supplement).

IV. RESULTS

Degradation (OE1). In the 10 baselines trained on ExDark with equal exposure, AP@0.5 goes from 0.61–0.69 unoccluded to 0.15–0.21 on occluded NightOwls people, from 0.47–0.65 to 0.08–0.15 on CityPersons *heavy* and from 0.78–0.85 to 0.20–0.31 on severe ExDark occlusion.

Global estimate (OE2–OE4). All figures are V3−V1 in AP@0.5 (Table I). All 20 controls pass, from +2.6 to +60.9 pp. With the usual dose and equal exposure (12 blocks, 114 per-seed differences), the mixed model estimates +0.57 [−0.34; +1.48] on NightOwls, +1.06 [+0.15; +1.96] on CityPersons

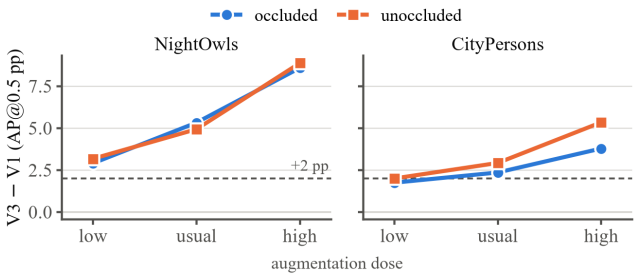


Fig. 2. Augmentation dose (H18, Faster R-CNN, silhouettes, 3 seeds): V3−V1 for occluded people (CityPersons *heavy*) and unoccluded people (*bare*). Dashed: +2 pp.

and −1.11 [−2.01; −0.20] on ExDark: all three upper limits lie below +2, so a mean gain of 2 pp is ruled out. Adding the block's gain on its control does not change the estimate (+0.16 pp per SD [−0.46; +0.78]). Descriptively, of those 36 cells 5 pass with the hierarchical CI, 9 rule out +2 pp and 22 are inconclusive; over all usual-dose blocks with a control, 5 of 54 pass.

Exposure. With unequal exposure the mean effect is −4.94 pp; the harm of silhouettes on NightOwls (−16.8 in H4) shrinks to −1.12 with fixed epochs (H10-s): exposure explains the harm, not the absence of gain.

Detector, domain and data (OE3, OE4). The pattern repeats with RT-DETR-l (H8), YOLO26s and YOLOv11m (H12). Faster R-CNN with silhouettes (H15) passes the rule on NightOwls (+3.28) and CityPersons (+2.33), also with the hierarchical CI ([+0.67; +5.81] and [+0.13; +4.92]). The 5-seed replication on another machine (H18a) keeps NightOwls

(+4.26 [+2.21; +6.32]) but not CityPersons (+1.74 [−0.08; +3.69]): 1 of 3. The tests that bound H15 do not transfer either: with diffusion and depth (H16a, H16b), Faster R-CNN learns the augmentation (+20.35 and +9.63) and no real cell exceeds +0.92, and RetinaNet with silhouettes (H17) gives 0 of 3, although its V3 worsens in every stratum. The deficit is not only due to the domain shift: within CityPersons (H6, H11), the controls gain +23.74 and +24.66 and *heavy* +1.82 and +1.31.

Dose (OE2). With Faster R-CNN, the gain grows with the dose on the control (+2.9, +14.8 and +25.6 pp), on NightOwls and on CityPersons, but not on ExDark (Fig. 2). The high dose passes on two sources, also with the hierarchical CI (H18e-h), but it is not specific to occlusion: unoccluded people gain as much or more (+8.87 and +5.34), and overall ExDark AP drops by 1.98.

Occluder (OE2). Neither diffusion, nor real geometry, nor depth meets the rule; in the cross-family matrix, 13 of 20 cells pass.

Post hoc analyses (OE5). *Without ExDark*, requiring NightOwls and CityPersons, only H15 and H18e-h detect transfer; at the usual dose, only H15, which is not replicated. *Specificity:* subtracting the gain on unoccluded people, of the 6 passing cells in H3–H17 only the two from the ExDark proxy are specific, and none withstands the hierarchical CI. *Visible box:* scoring CityPersons with the visible box, the *heavy* gain exceeds +1.3 in no block (H15: +0.74): part of what is gained is estimating the full extent of the person. *Realism* (H13, partially confirmed): with DINOv2 [**?**], each family's occlusion is told apart from real CityPersons occlusion (AUC 0.94–0.99) more easily than an unoccluded person (0.88).

V. Discussion

The results agree with Taori et al. [**?**] and Pytel et al. [**?**]: what is gained against a synthetic perturbation barely carries over to the real one. They do not contradict Kassaw et al. [**?**]: synthetic occlusion can rank models and still not suffice as a training signal. As in Hendrycks et al. [**?**] and Mintun et al. [**?**], carry-over depends on the augmentation–test pair and, here, on the dose. The high dose improves the external sources for visible and occluded people alike. Two readings explain this and neither is robustness to occlusion: regularization against domain shift, or more positives, because the high dose multiplies the pasted, labelled people by up to 3.5, like Copy-Paste augmentation [**?**]. That passing the control is not enough is compatible with a shortcut [**?**]: the artificial occluder remains distinguishable from the real one (H13), without a confirmatory test. For the 14,722 municipal cameras [**?**], the practical consequence is that a synthetic benchmark does not certify a model for night scenes with occluded people.

Limitations. *Scope:* 1,183 training images, fine-tuning from COCO and three doses tested only with Faster R-CNN and silhouettes; at another scale or dose the result may change. *Data:* almost all real occlusion is in datasets other than the training one; CityPersons is daytime, NightOwls is recorded from a vehicle and the ExDark proxy is post hoc, of low validity and $n = 101$; everything is single-class AP@0.5 at 640 px. *Protocol:* three seeds in most blocks; cells and blocks that share baselines, without multiplicity correction or dependence in the mixed model; unequal exposure in the first blocks; the mixed model is post hoc, and the fingerprint has no third-party timestamp (from now on it will be sealed with OpenTimestamps).

Future work. (1) A dose curve with other detectors, at larger scale and without labelling the pasted people. (2) Occlusion mined from real images and amodal supervision [**?**]. (3) Domain adaptation for night and part-based architectures [**?**].

VI. Conclusion

Real occlusion degrades the baseline in all three sources (OE1). Augmentation improved detection on synthetic occlusion in the 20 blocks with a control. On real occlusion, with the usual dose and equal exposure (12 blocks, four occluders, four detectors and two domains), the mixed model rules out a mean gain of 2 pp on all three sources (OE2–OE4); the only combination that passed the rule, Faster R-CNN with silhouettes, did not replicate with five seeds. A high dose does improve NightOwls and CityPersons, but equally for unoccluded people, so it is not robustness specific to occlusion. The synthetic occluder remains distinguishable from the real one (OE5). With pixel-level augmentation, 1,183 training images and single-class AP@0.5, passing a synthetic control is not enough to claim robustness to real occlusion: it takes annotated real occlusion, V1 and V3 on the same machine, five seeds and a specificity criterion.

Declarations

Data and code. Code, logs, fingerprints, sampling lists, the augmentation configuration, the mixed model, the cells with all their CIs and the supplement at https://github.com/jhanrq8888/argos-occlusion-transfer (public); trained weights will be deposited with a DOI on Zenodo upon publication. **Funding:** none. **Competing interests:** none. **Contributions (CRediT):** J. L. Ramos Quispe: conceptualization, methodology, software, formal analysis, investigation, data curation and writing of the original draft; C. P. Apaza Hilasaca: conceptualization, methodology (design of the pre-specification), review and editing; B. R. Condori Quispe: data curation, validation, visualization, review and editing. **Use of AI:** the authors used Scopus AI (Elsevier) to support the search for seed articles, and Claude (Anthropic) for the search, the analysis code and drafting and editing the text. They reviewed and verified the code, results, references and text, for which they take full responsibility.

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